History of Logic
Roman and Medieval Logic

Based on William and Martha Kneale, *The Development of Logic*, Chapter IV

Lecture Aim

The Roman and medieval period is not merely an interval between ancient logic and modern logic.

It begins with the transmission, translation, and partial fusion of the Aristotelian and Stoic traditions, but it eventually produces distinctively medieval theories of propositions, semantic reference, quantification, paradox, modality, and consequence.

The central historical movement of the lecture is:

TRANSMISSION $\longrightarrow$ RECOVERY $\longrightarrow$ INNOVATION

By the end of the period, medieval logicians have reconstructed a substantial body of primary, or propositional, logic and have begun to push logical theory beyond the boundaries of traditional Aristotelian syllogistic.

WHITEBOARD

Roman logic: preserve, translate, combine
↓
Medieval logic: recover, analyze, innovate

Main themes:

language
meaning and reference
quantification
paradox
consequence

Part I

From Cicero to Boethius

1 Two Ancient Traditions Become One

After Aristotle and Chrysippus, the Peripatetic and Stoic traditions did not remain permanently separate.

Later ancient writers discussed them together, borrowed terminology and doctrines from both, and gradually produced a mixed logical tradition.

The result was not always a clean synthesis.

The Kneales emphasize that what developed might be described as a fusion, but in some cases also as a confusion, of originally distinct systems.

The historical importance of this process is enormous because the mixed tradition becomes the logical inheritance passed to the Latin Middle Ages.

WHITEBOARD

ARISTOTLE
term / syllogistic logic

+

CHRYSIPPUS
propositional / Stoic logic

↓

late ancient debate and mixture

↓

BOETHIUS

fusion — or sometimes confusion

2 Cicero: Translator More Than Innovator

Cicero is not important here because he created a new formal system.

His importance lies primarily in transmission.

Greek logical terminology had to be rendered into Latin.

Cicero helped create a Latin philosophical vocabulary that later thinkers could inherit.

This matters because medieval logical work in western Europe would be conducted primarily in Latin.

One important example is the development of the word *propositio*.

Its later medieval meaning as a proposition or statement was not simply fixed from the beginning; Latin technical vocabulary developed historically.

WHITEBOARD

CICERO

Greek logical vocabulary ⟶ Latin logical vocabulary

↓

medieval logical language

Cicero: TRANSMISSION > INNOVATION

3 Topics and Formal Validity

Cicero's treatment of topics preserves an older distinction between two questions.

The first question is practical:

Where can I *find* an argument?

The second is logical:

Does the argument actually follow?

Topics are associated primarily with finding appropriate lines of argument.

Formal dialectic is concerned with whether a proposed inference is valid.

Cicero's interests remain closely connected with rhetoric and practical argumentation.

This reminds us that ancient and Roman logic is not yet always separated sharply from the art of persuasive discourse.

WHITEBOARD

TWO QUESTIONS

1. How do we FIND an argument?
   → TOPICS
2. Is the argument VALID?
   → DIALECTIC

Cicero emphasizes argument-discovery.

4 The Survival of Stoic Inferential Logic

The Stoic tradition did not disappear after Chrysippus.

Later ancient writers preserve schemes of hypothetical inference derived from, or related to, the earlier Stoic indemonstrables.

The evidence is imperfect, but a later tradition appears to have expanded and rearranged the basic Chrysippean material into a larger set of hypothetical rules.

The important historical point is not the exact reconstruction of every rule.

It is that propositional patterns of inference continued to circulate alongside Aristotelian syllogistic.

WHITEBOARD

Chrysippus

5 basic indemonstrables

↓

later expansion / rearrangement

↓

7 hypothetical rules

Stoic primary logic survives.

5 Galen: Two Logical Traditions

Galen explicitly works with material from both the Aristotelian and Stoic traditions.

Rather than treating one tradition as simply replacing the other, he recognizes that they have different characteristic applications.

Categorical syllogistic is naturally associated with geometrical reasoning.

Hypothetical reasoning is especially useful in metaphysical and dialectical contexts.

This is an important historical moment because the two systems are no longer merely rivals.

They can be understood as complementary logical tools.

WHITEBOARD

GALEN

CATEGORICAL LOGIC	HYPOTHETICAL LOGIC
↓	↓
geometry	metaphysical dialectic

We need BOTH kinds of reasoning.

6 Relational Reasoning: A Road Not Yet Taken

Galen also notices arguments whose validity depends on relations between objects.

Consider:

Theon has twice as much as Dio.

Philo has twice as much as Theon.

$\therefore$ Philo has four times as much as Dio.

Or:

Sophroniscus is father of Socrates.

$\therefore$ Socrates is son of Sophroniscus.

These inferences cannot be represented adequately merely by treating whole propositions as indivisible units, and they do not fit neatly into traditional subject–predicate syllogistic.

Their validity depends upon the internal logical structure of relations.

Galen sees the problem, but no systematic relational logic follows from the insight.

WHITEBOARD

$A = 2B$

$C = 2A$

$\therefore C = 4B$

and

X is father of Y

$\therefore Y$ is son of X

RELATIONAL reasoning
does not fit neatly into
ordinary syllogistic or basic Stoic forms

7 Porphyry and the Five Predicables

Porphyry's *Isagoge* becomes one of the central works of medieval logical education.

The famous five predicables are:

1. genus,
2. species,
3. differentia,
4. property,

5. accident.

These become known as the *quinque voces*.

They provide a framework for asking how predicates relate to subjects and how general concepts are organized.

WHITEBOARD

PORPHYRY: QUINQUE VOCES

GENUS
SPECIES
DIFFERENTIA
PROPERTY
ACCIDENT

Aristotle ⟶ Porphyry ⟶ medieval curriculum

8 Augustine and the *Dicibile*

Augustine's logical reflections distinguish among words, thoughts, and things.

The term *dicibile* is used for what is understood through a word and exists as intelligible content in the mind.

This is historically suggestive because it recalls the Stoic distinction between a linguistic expression and what that expression signifies.

The ancient problem of propositional content therefore remains alive, even though the technical vocabulary has changed.

WHITEBOARD

AUGUSTINE

WORD

↓

DICIBILE

"what is understood"

↓

thought / content

possible bridge back toward the Stoic *lekton*

9 Boethius: The Great Bridge

Boethius is the central transmitter of ancient logic to the early medieval Latin world.

His logical translations and commentaries preserve material from Aristotle, Porphyry, and the later ancient logical tradition.

For long periods, western students had access principally to works such as:

- Aristotle's *Categories*,
- Aristotle's *De Interpretatione*,
- Porphyry's *Isagoge*,
- Boethius' own logical works and commentaries.

Boethius is therefore not merely one author among many.

He is a historical conduit between Greek antiquity and Latin medieval thought.

WHITEBOARD

BOETHIUS

GREEK ANTIQUITY

↓

translations + commentaries

↓

LATIN MIDDLE AGES

BOETHIUS = THE BRIDGE

10 Boethius and Hypothetical Inference

Boethius helps preserve the tradition of hypothetical logic.

A representative pattern is:

$$A \to B,$$

$$B \to C,$$

therefore:

$$A \to C.$$

The point is historically important because the medieval West inherits not only categorical syllogistic but also remnants of propositional inference.

WHITEBOARD

HYPOTHETICAL TRANSITIVITY

$$\frac{\begin{array}{c} A \to B \\ B \to C \end{array}}{A \to C}$$

propositional material survives in Latin logic

11 Boethius and *Consequentia*

Boethius distinguishes between conditionals that hold merely accidentally and those that involve a more natural or necessary connection.

This helps preserve an important question:

What does it mean for one thing to *follow from* another?

The Latin vocabulary of *consequentia* will become increasingly important in medieval logic.

WHITEBOARD

CONSEQUENTIA

ACCIDENTAL

vs.

NECESSARY / NATURAL

"following from" becomes a central medieval problem

12 Written, Spoken, and Mental Discourse

The later ancient tradition distinguishes three kinds of discourse:

1. written,
2. spoken,
3. mental.

Written signs signify spoken signs.

Spoken expressions signify mental conceptions.

Mental discourse is treated as more fundamental and naturally connected with things.

This scheme becomes extremely important in medieval theories of mental language.

WHITEBOARD

THREE KINDS OF DISCOURSE

WRITTEN

↓

SPOKEN

↓

MENTAL

↓

THINGS

logic studies signs and what signs signify

13 The Problem of Universals

Porphyry raises questions about genera and species without resolving them.

Are universals:

- realities independent of thought?
- objects existing only in thought?
- corporeal or incorporeal?
- separate from sensible particulars or present only in them?

Boethius transmits these problems to medieval philosophy.

The problem of universals therefore becomes one of the major points at which logic, semantics, metaphysics, and epistemology intersect.

WHITEBOARD

THE UNIVERSALS PROBLEM

GENUS / SPECIES

things?
thoughts?
words?
incorporeal realities?

Porphyry asks ⟶ Boethius transmits ⟶ Middle Ages debate

Part II

From Alcuin to Abelard

14 The Early Medieval Logical Library

The early medieval West does not initially possess the whole Aristotelian logical corpus.

The principal inherited materials include:

- the *Categories*,
- *De Interpretatione*,
- Porphyry's *Isagoge*,
- the logical writings of Boethius.

These works later become associated with the *ars vetus*, the "old logic."

The limited textual inheritance is important because medieval thinkers begin rebuilding logical study from a comparatively narrow collection.

WHITEBOARD

EARLY MEDIEVAL LOGIC

Categories
De Interpretatione
Isagoge
Boethius

↓

ARS VETUS

15 Alcuin and the Trivium

Alcuin is important less for original logical theory than for education.

The traditional trivium is:

1. grammar,
2. dialectic,
3. rhetoric.

Dialectic thereby becomes one of the basic disciplines of higher intellectual training.

WHITEBOARD

TRIVIUM

GRAMMAR
DIALECTIC
RHETORIC

logic becomes structurally embedded in education

16 Grammar and Logic

The recovery of logic occurs alongside intense study of grammar.

This creates both opportunities and confusions.

Medieval thinkers repeatedly ask how to distinguish:

- the physical or written word,
- the meaning of the word,
- the thing named,
- the universal associated with the term.

These questions gradually become logical and semantic problems in their own right.

WHITEBOARD

MEDIEVAL SEMANTIC PROBLEM

WORD

?

MEANING

?

THING

Grammar + Logic
can illuminate one another
or confuse one another

17 Nominalism and Realism

The proposition

Homo est species

means:

Man is a species.

But what does the subject term stand for?

If a species is something real, we move toward realism.

If the universal is only a word or sound, we move toward nominalism.

Roscellin becomes associated with a strong nominalist tendency according to which universals are merely words or a *flatus vocis*.

Abelard rejects simple realism but also does not accept a crude theory in which universals are nothing but noises.

WHITEBOARD

UNIVERSALS

REALISM
universal has reality

vs.

NOMINALISM
universal = word / *flatus vocis*

ABELARD: neither simple extreme

18 A New Intellectual Confidence

A major transformation occurs in the twelfth century.

The goal is no longer merely to recover and preserve the ancients.

Thinkers increasingly assume that ancient philosophy can be analyzed, criticized, and improved.

This change in intellectual confidence is essential for understanding Abelard.

WHITEBOARD

EARLY MIDDLE AGES

recover the ancients

↓

TWELFTH CENTURY

argue with the ancients

↓

improve the ancients

19 Anselm and Dialectical Theology

Anselm applies dialectical reasoning directly to theological questions.

One important method is reductive reasoning.

Suppose the denial of a proposition.

If that supposition produces an impossibility or contradiction, reject the denial.

Schematically:

$$\neg P \rightarrow \bot$$

therefore:

$$P.$$

The important point for the history of logic is methodological: logical patterns inherited from antiquity are now being used systematically in theological argument.

WHITEBOARD

ANSELM

suppose $\neg P$

$\downarrow$

derive contradiction

$\downarrow$

$\boxed{P}$

ancient dialectical forms $\rightarrow$ medieval theology

20 The Scholastic *Quaestio*

Abelard's *Sic et Non* exemplifies a style of inquiry that becomes characteristic of scholastic intellectual life.

One begins with apparently conflicting claims or authorities.

The task is not simply to choose one side immediately.

Instead one must identify distinctions that allow the apparent contradiction to be analyzed.

WHITEBOARD

SCHOLASTIC QUAESTIO

PRO

vs.

CONTRA

$\downarrow$

DISTINCTIONS

$\downarrow$

$\boxed{\text{RESOLUTION}}$

Reason advances through organized disputation.

21 Abelard as an Innovator

Abelard is not merely a compiler.

His logical work shows confidence that the ancient tradition is incomplete and that further discoveries can be made.

His *Dialectica* therefore marks an important transition from recovery to active development.

WHITEBOARD

ABELARD

Ancient logic $\neq$ finished logic

Logic remains a field of discovery.

22 What Is True or False?

Abelard defines a proposition as discourse signifying truth or falsity.

Yet he recognizes that the fundamental concern of logic cannot be merely the physical utterance or written sentence.

A proposition is logically interesting because of what it puts forward as being the case.

Thus we need to distinguish the propositional sign from what is expressed by the sign.

WHITEBOARD

ABELARD

PROPOSITION-SIGN

↓

expresses

↓

WHAT IS THE CASE

↓

TRUE / FALSE

truth concerns propositional content

23 The *Dictum*

Abelard uses the term *dictum* for what is expressed by a proposition.

For example:

Socrates is a man

expresses:

Socrates' being-a-man.

But the *dictum* is not simply another ordinary object alongside Socrates and other things.

The distinction is one of the important medieval attempts to understand propositional content.

WHITEBOARD

"Socrates is a man"

↓

DICTUM

Socrates' being-a-man

what is expressed ≠ another ordinary thing

24 The Copula

Abelard gives special logical attention to the verb *est*.

In a proposition such as:

Socrates is a man

the word "is" functions as a copula joining subject and predicate.

It should not automatically be understood as an independent assertion of existence.

This distinction is important because otherwise every use of the copula would appear to commit the speaker to the existence of the subject in a way that ordinary predication does not.

WHITEBOARD

Socrates **IS** a man

↑

COPULA

copula ≠ independent assertion of existence

25 Toward Supposition Theory

A central medieval question becomes:

What does a term stand for in a proposition?

Abelard's semantic analysis helps prepare the way for the later theory of *suppositio terminorum*.

The same word can perform different logical roles in different contexts.

The later medievals will turn this insight into an elaborate technical theory.

WHITEBOARD

TERM IN A PROPOSITION

What does it stand for HERE?

ABELARD

↓

SUPPOSITIO TERMINORUM

26 Negation Has Scope

Abelard sees that negation can apply to a whole proposition rather than merely to one term inside it.

Compare:

Not every man is white

with:

Some man is not white.

The two should not simply be treated as identical without examining assumptions about existence and quantification.

This is a major step toward explicit analysis of logical scope.

WHITEBOARD

SCOPE OF NEGATION

¬[Every man is white]

$\neq$ automatically

Some man is ¬ white

negation of the WHOLE
must be distinguished from
negation INSIDE the proposition

27 Existential Import and the Square

Abelard worries about the traditional square of opposition because he takes universal affirmative propositions to carry existential import.

Consider:

Every man is a man.

If there are no men, should this still count as true?

Abelard's answer is no.

This shows that the traditional relations among quantified propositions contain assumptions about existence that must be made explicit.

WHITEBOARD

EXISTENTIAL IMPORT

Every man is a man.

Suppose: NO MEN EXIST

True or false?

ABELARD: FALSE

The traditional square contains existential assumptions.

28 Quantifiers as Logical Operators

Abelard explicitly analyzes the signs of quantity:

omnis, *quidam*, *nullus*.

That is:

every, some, no.

The logical form of a proposition depends in part on how these expressions determine the range of the subject term.

WHITEBOARD

SIGNA QUANTITATIS

OMNIS → EVERY
QUIDAM → SOME
NULLUS → NO

logical form depends on quantifier behavior

29 Modal Scope

Abelard also distinguishes two ways modal language can operate.

Compare:

It is possible that no man is white

with:

No man is possibly white.

In the first case, possibility applies to an entire proposition.

In the second, the modal expression appears within the predication.

This anticipates the later distinction between modality *de dicto* and modality *de re*.

WHITEBOARD

MODAL SCOPE

POSSIBLE [No man is white]

$\neq$

No man is [POSSIBLY white]

later terminology:
DE DICTO vs. *DE RE*

30 Abelard's Theory of Consequence

For Abelard, a satisfactory conditional expresses a necessary connection.

He speaks of *necessitas consecutionis*.

He distinguishes arguments whose validity is secured by their formal arrangement from arguments that require information about the particular things denoted by their terms.

The contrast can be expressed as structure versus special facts about the terms.

WHITEBOARD

ABELARD: CONSEQUENCE

FORMAL / PERFECT
truth from STRUCTURE

vs.

IMPERFECT
needs facts about TERMS / THINGS

31 The Paradox of Implication

Suppose the antecedent of a conditional is impossible.

For example:

Socrates is a stone.

Then the combination:

Socrates is a stone and Socrates is not an ass

is itself impossible.

This makes it tempting to say:

If Socrates is a stone, then Socrates is an ass.

But Abelard resists treating such a conditional as expressing a genuine intrinsic connection.

For strict consequence, the antecedent should require the consequent *ex se ipso*, from itself.

WHITEBOARD

PARADOX OF IMPLICATION

Impossible P

Does $P \to Q$

hold for ANY Q?

ABELARD: not in the strict sense
Antecedent must require consequent intrinsically

32 Abelard's Inferential Rules

Abelard explicitly works with several familiar patterns.

Modus Ponens

$$P \to Q$$

$$P$$

$$\therefore Q$$

Modus Tollens

$$P \to Q$$

$$\neg Q$$

$$\therefore \neg P$$

Transitivity

$$P \to Q$$

$$Q \to R$$

$$\therefore P \to R$$

WHITEBOARD

ABELARD

$$\frac{P \to Q \qquad P}{Q}$$

$$\frac{P \to Q \qquad \neg Q}{\neg P}$$

$$\frac{P \to Q \qquad Q \to R}{P \to R}$$

modus ponens
modus tollens
transitivity

Part III

The Logic of the Universities

33 The *Ars Nova*

By the middle of the twelfth century, much more of Aristotle's logical corpus becomes available in Latin.

The newly recovered portions of the *Organon* become known as the *ars nova*.

The older core remains the *ars vetus*.

The result is a greatly expanded logical curriculum.

WHITEBOARD

ARS VETUS

+

ARS NOVA

↓

UNIVERSITY LOGIC

34 Logic Becomes a University Discipline

Logic becomes central to the arts curriculum at universities such as Paris and Oxford.

Its characteristic forms include:

- lectures,
- commentaries,
- *quaestiones*,
- disputations,
- handbooks and *summulae*.

Logical reasoning is no longer merely something read in ancient texts.

It is a practiced academic discipline.

WHITEBOARD

MEDIEVAL UNIVERSITY LOGIC

LECTURE

↓

QUAESTIO

↓

DISPUTATION

↓

FORMAL TRAINING

35 Sophisms as Logical Laboratories

The recovery of Aristotle's *De Sophisticis Elenchis* stimulates an enormous interest in sophisms.

A *sophisma* can be used as a deliberately difficult case.

The point is not merely to avoid mistakes.

The puzzle forces the logician to identify ambiguity, scope, reference, and inference patterns.

Sophisms therefore function as intellectual laboratories.

WHITEBOARD

SOPHISMA

construct a difficult case

↓

force distinctions

↓

test logical theory

logical puzzle = laboratory for analysis

36 *Insolubilia*: The Liar Returns

Variants of the Liar paradox become known as *insolubilia*.

A simple example is:

"This proposition is false."

The medievals recognize that the difficulty involves a peculiar kind of self-reference.

Different solutions are proposed, including:

- *cassatio*,
- *restrictio*,
- solution *secundum quid et simpliciter*.

Cassatio attempts to deny that the problematic expression successfully states anything.

Restrictio restricts the range of reference in order to block the vicious self-reference.

WHITEBOARD

INSOLUBILIA

"This proposition is false."

CASSATIO
RESTRICTIO
SECUNDUM QUID / SIMPLICITER

self-reference becomes a technical logical problem

37 *Syncategoremata*

Medieval logicians write special treatises on expressions that do not ordinarily function as terms but are essential to logical structure.

Examples include:

and, or, not, if, every, some, only, except.

These are *syncategoremata*.

The important insight is that logical form can be controlled by very small pieces of language.

WHITEBOARD

SYNCATEGOREMATA

AND
OR
NOT
IF
EVERY
SOME
ONLY
EXCEPT

small words — large effects on LOGICAL FORM

38 *Obligationes*

An *obligatio* is a regulated disputational situation.

A participant agrees to an assumption or set of conditions and must then answer subsequent questions consistently within that framework.

This turns logical disputation into a rule-governed exercise.

It continues the ancient interest in dialectical debate but with a more elaborate formal structure.

WHITEBOARD

OBLIGATIONES

assume P

↓

conduct dispute

↓

maintain consistency

logic as a rule-governed dialectical game

39 William of Shyreswood and Peter of Spain

William of Shyreswood's *Introductiones in Logicam* organizes material on:

- propositions,
- predicables,
- syllogisms,
- dialectical topics,
- properties of terms,
- fallacies.

Peter of Spain's *Summulae Logicales* becomes an even more influential logical handbook.

The university manuals combine inherited Aristotelian doctrine with specifically medieval developments.

WHITEBOARD

HIGH MEDIEVAL MANUALS

William of Shyreswood

↓

Peter of Spain

↓

SUMMULAE tradition

ancient logic + medieval innovations

40 Mental Language

The theory of written, spoken, and mental discourse receives new development under the influence of Aristotle and Arabic philosophy.

Mental terms are treated as *intentiones*, natural signs or concepts in the soul.

This creates a theory in which conventional spoken and written signs are subordinated to a more fundamental mental language.

WHITEBOARD

MENTAL LANGUAGE

WRITTEN SIGNS

↓

SPOKEN SIGNS

↓

MENTAL SIGNS

↓

THINGS

intentio = mental sign / concept

41 Ockham's Systematic Logic

William Ockham attempts to organize the entire subject of logic.

His *Summa Totius Logicae* is divided broadly into:

1. terms,

2. propositions,

3. arguments.

Within this framework he incorporates both traditional logical material and specifically medieval developments.

WHITEBOARD

OCKHAM

BOOK I → TERMS
BOOK II → PROPOSITIONS
BOOK III → ARGUMENTS

ancient inheritance + medieval logical theory

42 The Fourteenth-Century Logical Culture

After Ockham, important figures include:

- Walter Burleigh,
- Jean Buridan,
- Richard Swineshead,
- Albert of Saxony,
- Marsilius of Inghen,
- William Heytesbury,
- Ralph Strode,
- Richard Ferrybridge.

This is a period of exceptionally intense logical activity.

Medieval thinkers continue refining the theories of terms, consequences, modality, sophisms, paradox, and disputation.

WHITEBOARD

FOURTEENTH-CENTURY LOGIC

Burleigh
Buridan
Swineshead
Albert of Saxony
Heytesbury
Strode
others

one of the most intensive periods of logical study

43 *Via Antiqua* and *Via Moderna*

The later medieval intellectual world develops competing educational tendencies.

The *via antiqua* is associated with older Aristotelian and realist approaches.

The *via moderna* is associated more strongly with terminist and nominalist developments and with newer kinds of logical analysis.

The difference eventually becomes institutional.

WHITEBOARD

VIA ANTIQUA

realism + older Aristotelian emphasis

vs.

VIA MODERNA

terminism / nominalist tendency

newer logical analysis

Part IV

Proprietates Terminorum

44 The Properties of Terms

The theory of the *proprietates terminorum* develops in the later twelfth century and becomes one of the most distinctive areas of medieval logic.

The basic question is:

What logical role does a word or phrase perform when it occurs as a term in a proposition?

William of Shyreswood gives four principal properties:

1. *significatio*,
2. *suppositio*,
3. *copulatio*,
4. *appellatio*.

WHITEBOARD

PROPRIETATES TERMINORUM

SIGNIFICATIO
SUPPOSITIO
COPULATIO
APPELLATIO

What is a term doing IN THIS PROPOSITION?

45 *Significatio* and *Suppositio*

A useful introductory distinction is:

significatio

concerns what a term means or signifies.

suppositio

concerns what the term stands for in a particular propositional context.

The same expression can retain a general meaning while standing for different kinds of things in different occurrences.

WHITEBOARD

SIGNIFICATIO

"What does the word MEAN?"

SUPPOSITIO

"What does it stand for HERE?"

meaning ≠ contextual reference

46 Three Core Types of Supposition

The full medieval taxonomies are extremely elaborate.

For introductory purposes, three central cases are especially useful.

Material Supposition

The expression stands for the linguistic sign itself.

Example:

"Man" is a word.

Simple Supposition

The expression stands for a universal, concept, or associated mental sign.

Example:

Man is a species.

Personal Supposition

The expression stands for the individual things falling under the term.

Example:

Every man is mortal.

WHITEBOARD

THREE CORE CASES

1. MATERIAL
   "man" as linguistic sign
2. SIMPLE
   man as concept / universal
3. PERSONAL
   individual men

same term — different logical role

47 Personal Supposition and Quantification

Within personal supposition, medieval thinkers distinguish different ways in which a common term ranges over individuals.

A simplified teaching scheme is:

- determinate,
- confused,
- distributive.

These distinctions function partly as a substitute for what modern logic handles through explicit quantifier notation.

WHITEBOARD

PERSONAL SUPPOSITION

DETERMINATE
→ some particular member

CONFUSED
→ range not individually fixed

DISTRIBUTIVE
→ all relevant members

medieval substitute for parts of quantification theory

48 Ockham's Nominalist Revision

William of Shyreswood and Peter of Spain work within a framework in which a general term can be said to signify a common form or universal.

Ockham rejects the existence of such a universal entity.

For him, signs ultimately signify individual things.

This makes *suppositio* increasingly central to semantic theory.

WHITEBOARD

WILLIAM / PETER

general term → common form / universal

vs.

OCKHAM

general term → individual things

suppositio becomes more fundamental

49 Burleigh's Objection

Walter Burleigh objects to Ockham's account of signification.

Suppose the word "man" directly meant Socrates, Plato, and the other individual men.

Then learning the meaning of the word "man" should require learning that the word applies to those individuals.

But someone can understand the word "man" without knowing Socrates.

Therefore the meaning of a general term cannot simply be identified with a list of individuals.

WHITEBOARD

BURLEIGH AGAINST OCKHAM

If "man" meant

Socrates + Plato + $\cdots$

then learning "man"

would require knowing Socrates.

But it does not.

MEANING $\neq$ list of individuals

50 Distribution

The doctrine of distribution is the most familiar surviving part of medieval term theory.

A term is distributed when it is taken for all the individuals to which it applies.

For example:

Every man is mortal.

The subject term "man" is taken universally.

Later traditional syllogistic develops two famous rules:

1. The middle term must be distributed at least once in the premises.
2. A term may not be distributed in the conclusion unless it was distributed in a premise.

WHITEBOARD

DISTRIBUTION

Every MAN is mortal.

MAN → taken for every man

1. Middle term distributed at least once.
2. No new distribution in the conclusion.

51 What Term Theory Was Trying to Do

The theory of the properties of terms combines several different projects.

First, it attempts a general theory of language involving meaning, application, and reference.

Second, it analyzes the detailed behavior of the natural language used by philosophers.

Third, it develops rules for quantified inference.

These are all respectable projects, but they do not necessarily belong inside one single theory.

WHITEBOARD

PROPRIETATES TERMINORUM

1. theory of LANGUAGE
   meaning / reference
2. analysis of logical IDIOMS
3. rules of QUANTIFICATION

valuable projects — but not naturally one theory

52 The Ambiguity of *Suppositio*

The basic distinction between signification and supposition is useful.

The difficulty is that the word *suppositio* becomes overloaded.

Compare:

"Socrates" stands for Socrates

with:

"man" stands for Socrates, Plato, and others.

These are not obviously the same semantic relation.

Yet both can be described using the language of "standing for."

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PROBLEM WITH *SUPPOSITIO*

1. A name → Socrates

2. MAN → Socrates, Plato, ...

Both called: "stands for"

but these are different semantic relations

Part V

Consequentiae

53 What Is a *Consequentia*?

The later medieval theory of *consequentiae* develops from earlier work associated with Abelard, although the exact historical chain is not completely known.

The central question is:

Under what conditions does one proposition validly follow from another?

This broadens logical investigation beyond the traditional theory of categorical syllogism.

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CONSEQUENTIA

P

$\downarrow$

Q

WHEN DOES Q FOLLOW FROM P?

54 Two Paradoxes of Consequence

Two medieval principles can appear surprising.

First:

anything follows from the impossible.

In modern notation:

$$\bot \vdash P.$$

Second:

the necessary follows from anything.

If N is necessary, then:

$$P \vdash N.$$

The intuitive explanation concerns the impossibility of a counterexample in which the antecedent is true and the consequent false.

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PARADOXES OF CONSEQUENCE

IMPOSSIBLE

↓

ANYTHING

and

ANYTHING

↓

NECESSARY TRUTH

Can there be a true antecedent and false consequent?

55 Formal and Material Consequence

A particularly important medieval distinction separates formal from material consequence.

A consequence is formal when its validity remains intact under suitable replacement of its non-logical terms while the logical structure is preserved.

A material consequence depends upon the special meanings of the terms or upon additional facts.

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FORMAL CONSEQUENCE

valid under replacement

while FORM stays fixed

vs.

MATERIAL CONSEQUENCE

valid because of particular meanings / facts

56 Substitution as a Test of Form

Consider:

Every M is L.

Every S is M.

$\therefore$ Every S is L.

Now replace M, L, and S with different suitable terms.

If the argument remains valid under the substitutions, then the validity belongs to the structure rather than the specific subject matter.

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FORMAL TEST

$$\frac{\begin{array}{c}\text{Every } M \text{ is } L \\ \text{Every } S \text{ is } M\end{array}}{\text{Every } S \text{ is } L}$$

replace M, L, S
validity survives
$\Rightarrow$ FORM carries the inference

57 *Bona Simpliciter* and *Bona Ut Nunc*

Medieval logicians also distinguish kinds of material consequence.

A consequence can be good *simpliciter*, because a stable background relation between the terms supplies what is needed.

Or it can be good *ut nunc*, "as things now stand," because some contingent circumstance currently makes the inference work.

Suppose:

Socrates is white.

Then:

Socrates runs

may support:

A white thing runs.

But this works because of the contingent fact that Socrates is white.

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TWO MATERIAL CASES

BONA SIMPLICITER
stable background truth

vs.

BONA UT NUNC
true given present circumstances

Socrates is white

Socrates runs

$\therefore$ A white thing runs

58 Ockham's General Principles

Ockham gives a large set of rules governing consequence.

For lecture purposes, the following family captures the central ideas:

1. A true proposition does not validly entail a false proposition.
2. If P entails Q, then the negation of Q entails the negation of P.
3. If P entails Q, and Q entails R, then P entails R.
4. Anything follows from the impossible.
5. A necessary truth follows from anything.

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OCKHAM: GENERAL PRINCIPLES

1. TRUE does not entail FALSE
2. $P \vdash Q \Rightarrow \neg Q \vdash \neg P$
3. $P \vdash Q,\ Q \vdash R \Rightarrow P \vdash R$
4. $\bot \vdash P$
5. $P \vdash N \quad (N \text{ necessary})$

59 Rules Later Associated with De Morgan

Medieval logicians formulate rules corresponding to what later became known as De Morgan's laws.

For conjunction:

$$\neg(P \wedge Q) \leftrightarrow (\neg P \vee \neg Q).$$

For disjunction:

$$\neg(P \vee Q) \leftrightarrow (\neg P \wedge \neg Q).$$

The nineteenth-century name is much later than the logical insight.

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NEGATION RULES

$$\neg(P \wedge Q) \leftrightarrow (\neg P \vee \neg Q)$$

$$\neg(P \vee Q) \leftrightarrow (\neg P \wedge \neg Q)$$

later name: De Morgan
medieval rules: centuries earlier

60 Primary Logic Re-emerges

The medieval theory of consequences contains rules involving:

- conjunction,
- disjunction,
- negation,
- conditionals,
- contraposition,
- transitivity.

This is precisely the territory previously identified as primary or propositional logic.

The tradition therefore reconstructs a substantial portion of the logical domain earlier developed by the Stoics.

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MEDIEVAL *CONSEQUENTIAE*

$\neg$
$\wedge$
$\vee$
$\rightarrow$
contraposition
transitivity

PRIMARY / PROPOSITIONAL LOGIC

61 Beyond Syllogistic: Existential Inference

Medieval writers also encounter forms of inference that move beyond traditional syllogistic.

For example:

Socrates runs.

Therefore:

Something runs.

In modern notation:

$$R(s)$$

therefore:

$$\exists x\, R(x).$$

This is a direct connection between singular predication and existential quantification.

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Socrates runs

$$\downarrow$$

Something runs

$$\frac{R(s)}{\exists x\, R(x)}$$

general logic begins to exceed ordinary syllogistic

62 The Limits of Medieval Notation

Medieval logicians investigate many difficult topics:

- relations,
- quantification,
- scope,
- modal contexts,
- semantic paradox,
- formal replacement,
- propositional connectives.

But they do not possess a compact symbolic language adequate to express all of these structures with the clarity later provided by modern quantificational notation.

The limitation is therefore not simply lack of intelligence or logical sophistication.

They repeatedly confront problems whose solution requires a representational advance.

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MEDIEVAL LOGICIANS HAD

logical insight

+

technical vocabulary

BUT LACKED

adequate symbolism for quantification

∴

further progress becomes difficult

63 Comparison with Chrysippus

The later medieval writers can derive one valid consequence from another and use higher-order principles such as contraposition.

In that respect their work recalls Chrysippus, who derived further valid schemata from his indemonstrables.

But no surviving medieval work appears to construct a single highly developed deductive calculus of consequences comparable to a modern formal system.

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CHRYSIPPUS

basic schemata

↓

derived schemata

MEDIEVAL *CONSEQUENTIAE*

valid consequences

↓

derive further consequences

but no single unified modern-style calculus

Part VI

Synthesis

64 What Changed from Rome to the High Middle Ages?

The Roman period preserves and combines ancient traditions.

The early Middle Ages recover a restricted logical inheritance.

The twelfth century brings renewed confidence and the work of Abelard.

The universities then institutionalize logical training and generate specialized technical traditions.

Two developments become especially characteristic:

1. the theory of the properties of terms,
2. the theory of consequences.

The first pushes toward semantics and quantification.

The second pushes toward a general theory of propositional inference.

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ROMAN PERIOD

PRESERVE TRANSLATE MIX

↓

EARLY MIDDLE AGES

RECOVER

↓

ABELARD

QUESTION DISTINGUISH INNOVATE

↓

HIGH MIDDLE AGES

SEMANTICS
QUANTIFICATION
PARADOX
CONSEQUENCE

65 Worked Board Demonstrations

1. Relational Inference

$$A = 2B, \qquad C = 2A.$$

Therefore:

$$C = 4B.$$

The validity depends on relations internal to the propositions.

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$$A = 2B$$

$$C = 2A$$

$$\therefore C = 4B$$

RELATIONAL STRUCTURE matters

2. The Copula

Consider:

A chimaera is imaginary.

The word "is" joins subject and predicate.

It does not by itself assert that a chimaera exists as an ordinary object.

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A chimaera IS imaginary

↑

COPULA

"is" does not automatically mean "exists"

3. Negation and Quantification

Compare:

$$\neg(\forall x(M(x) \rightarrow W(x)))$$

with:

$$\exists x(M(x) \wedge \neg W(x)).$$

In modern logic they are equivalent under the standard interpretation of the quantifiers.

But Abelard's historical problem concerns the existential assumptions built into older logical doctrine.

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$\neg$[Every S is P]

vs.

Some S is not P

scope + existential assumptions must be examined

4. Modal Scope

Compare:

$$\Diamond\neg\exists x(M(x) \wedge W(x))$$

with:

$$\forall x(M(x) \rightarrow \Diamond\neg W(x)).$$

The position of the modal operator changes what is asserted.

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POSSIBLE [No S is P]

$\neq$

No S is [POSSIBLY P]

modal scope changes meaning

5. Paradox of Implication

Suppose P is impossible.

Then no case exists in which P is true and Q false.

Does that establish:

$$P \rightarrow Q$$

for any Q?

That question separates a truth-conditional conception of implication from Abelard's stronger demand for intrinsic connection.

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$P =$ impossible

Does $P \rightarrow Q$

hold for arbitrary Q?

truth condition vs. intrinsic connection

6. One Word, Three Suppositions

Use the same expression in three different contexts.

"man" is a noun

man is a species

every man is mortal

The logical role of the expression changes.

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"man"

1. linguistic sign
2. species / concept
3. individual men

same expression — different supposition

7. Formal Consequence by Substitution

Begin with:

Every M is L.

Every S is M.

$\therefore$ Every S is L.

Replace the terms repeatedly.

The validity remains.

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$$\frac{\text{Every } M \text{ is } L \qquad \text{Every } S \text{ is } M}{\text{Every } S \text{ is } L}$$

substitution preserves validity

8. A Small Medieval Rule Set

A compact group of rules displays how much propositional structure the medieval tradition had recovered.

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$$P \wedge Q \vdash P$$

$$P \vdash P \vee Q$$

$$P \to Q,\ P \vdash Q$$

$$P \to Q,\ \neg Q \vdash \neg P$$

$$\neg(P \wedge Q) \leftrightarrow (\neg P \vee \neg Q)$$

substantial PRIMARY LOGIC has re-emerged

66 Final Summary

The Roman and medieval history of logic should not be described merely as a period of decline between Greek antiquity and the Renaissance.

The Roman tradition preserves ancient logical material and makes it available in Latin.

Boethius becomes the decisive bridge.

The early medieval world inherits only part of ancient logic, but the twelfth century sees renewed confidence and original logical work.

Abelard makes major advances in the analysis of propositions, propositional content, copulation, negation, modality, quantification, and consequence.

The medieval universities transform logic into a large-scale academic discipline.

Their interest in sophisms, *insolubilia*, *syncategoremata*, and *obligationes* makes difficult linguistic and inferential cases into subjects of sustained technical analysis.

The theory of the *proprietates terminorum* attempts to explain meaning, reference, and quantificational behavior.

The theory of *consequentiae* broadens the theory of valid inference and recovers a large part of what can be recognized as primary or propositional logic.

Medieval thinkers do not yet create modern quantification theory.

A major obstacle is the absence of a sufficiently powerful symbolic notation.

Nevertheless, they identify many of the problems that modern logic will later solve in a more systematic symbolic language.

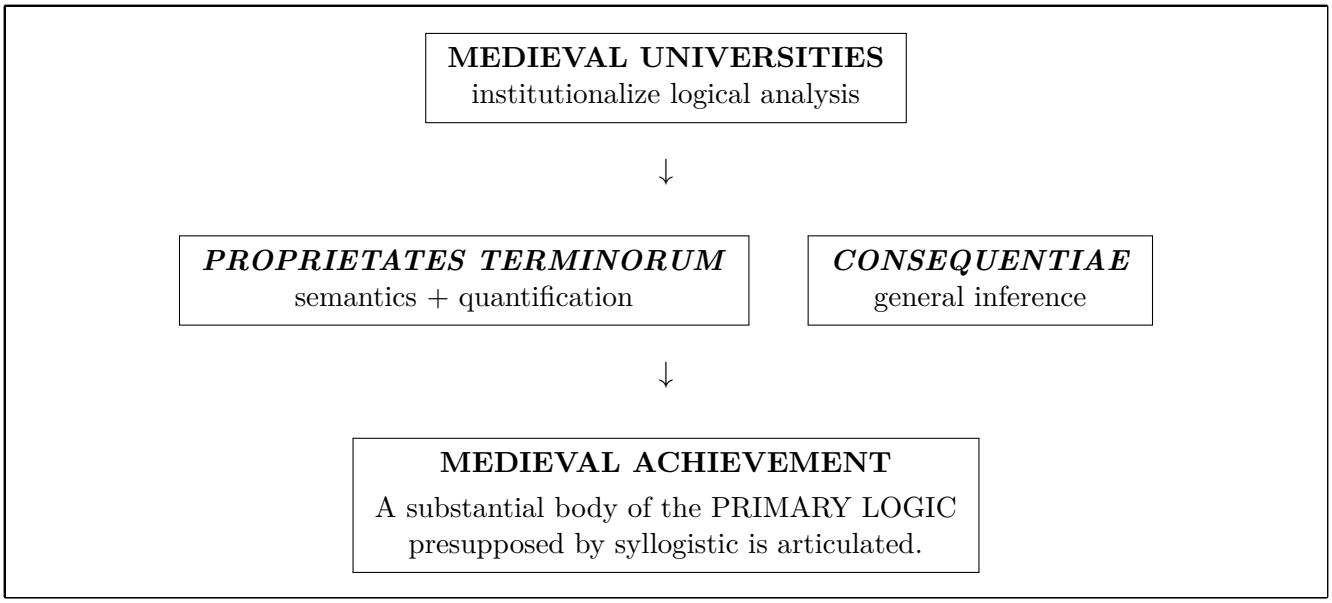
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ROMAN LOGIC
preserves and transmits

↓

ABELARD
turns recovery into innovation

↓



Master Whiteboard Packet

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1. Ancient inheritance

Aristotle + Chrysippus → late ancient mixture → Boethius

2. Early medieval recovery

ars vetus → trivium → universals controversy

3. Abelard

proposition → *dictum* → copula

negation → quantification → modal scope

consequence → modus ponens / tollens / transitivity

4. University logic

ars nova → sophismata → *insolubilia*

syncategoremata → *obligationes*

5. Term theory

significatio ≠ *suppositio*

material simple personal

distribution → quantificational analysis

6. Theory of consequences

formal vs. material

substitution → formal validity

$\neg,\ \wedge,\ \vee,\ \rightarrow$

PRIMARY / PROPOSITIONAL LOGIC

7. Historical limit

strong logical analysis + weak notation

↓

difficulty developing full quantification theory

8. Final historical trajectory

TRANSMISSION → RECOVERY → INNOVATION